HM Government

T-LEVELS

Student Workbook: Principles of Collision Theory

T Level Technical Qualification in Science • Chemistry • A11.1

Name: _____ Date: _____

Specification coverage

Point	Required understanding
A11.1.1	Particles must collide with enough energy to break and reform bonds (activation energy).
A11.1.2	Particles must be in the correct spatial orientation.

1. Retrieval starter

1.1 Initial thinking

Why might two reactant particles collide without producing products? Give two possible reasons.

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2. Activation energy

2.1 Definition

Define activation energy.

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2.2 Particle explanation

Explain why collision energy below the activation energy does not produce the specified reaction.

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Collision	Activation energy / kJ mol^{-1}	Collision energy / kJ mol^{-1}	Energy requirement met?	Reason
A	45	30		
B	45	45		
C	45	52		
D	62	58		

2.3 Bond changes

Explain why energy is required when bonds must break and reform during a reaction.

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3. Spatial orientation

3.1 Meaning

Explain what 'correct spatial orientation' means in collision theory.

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3.2 Orientation reasoning

A collision has more than the activation energy but the reacting parts of the particles do not meet. Predict the outcome and explain.

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4. Applying both conditions

Scenario	Ea / kJ mol^{-1}	Collision energy / kJ mol^{-1}	Orientation	Successful?	Justification
1	45	30	Correct		
2	45	55	Incorrect		
3	45	50	Correct		
4	62	70	Correct		
5	62	61	Correct		

4.1 Rule

Write a two-step decision rule for deciding whether a collision is successful.

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5. Error analysis

5.1

‘Every collision produces a reaction.’ Correct the statement.

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5.2

'If collision energy is high enough, orientation does not matter.' Correct the statement.

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5.3

'Correct orientation can make up for collision energy below activation energy.' Correct the statement.

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6. Unfamiliar collision challenge

6.1 Collision X

Reaction X has Ea = 80 kJ mol^{-1}. A collision has 92 kJ mol^{-1} and incorrect orientation. Decide whether it is successful and justify.

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6.2 Collision Y

Reaction Y has Ea = 35 kJ mol^{-1}. A collision has 35 kJ mol^{-1} and correct orientation. Decide whether it can be successful and justify.

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7. Exam style questions

7.1 [2]

A collision has 58 kJ mol^{-1}. The activation energy is 62 kJ mol^{-1} and orientation is correct. Explain why the collision is unsuccessful.

7.2 [2]

A collision has 70 kJ mol^{-1}. The activation energy is 62 kJ mol^{-1} but orientation is incorrect. Explain why the collision is unsuccessful.

7.3 [2]

State the two conditions required for a successful collision.

7.4 [2]

Explain why many collisions between reactant particles do not form products.

8. Final knowledge check

8.1

Define activation energy.

8.2

What happens if collision energy is below activation energy?

8.3

What is correct spatial orientation?

8.4

Can sufficient energy alone guarantee reaction? Explain.

8.5

State both requirements for a successful collision.